Name:_____ **Period:_____**

Introductory ESS Language Decoder Quiz Study Guide

This guide covers the 25 vocabulary terms tested on the quiz, organized into the same three parts as the quiz. For each term: the definition, the "think" question to help you recognize it, and an example from the activities.

The quiz format: you'll be given a scenario or question stem (not the definition itself) and have to pick the vocabulary word that best matches it. Practice by re-reading the activity examples above and asking "which term describes what's happening here?"

Part 1: Dimensions and Scale (5 terms)

Term	Definition	Think:	Example
Spatial	Used to describe changes, differences, or patterns in physical space or geographic location.	Where?	The bath toys' landfall locations tell us WHERE they were found.
Relative	Used to compare one value, amount, size, or position to another. Does not necessarily give an exact measurement.	Compared to what?	The Great Pacific Garbage Patch's location described compared to the Ever Laurel accident location.
Approximate / Approximately	Indicates an estimated value, amount, distance, or time that is close to, but not exactly, the actual value.	About how much?	The Garbage Patch is approximately 11 times the size of New York State.
Temporal	Refers to time, changes over time, or the timing of specific events.	When? How long? Over time?	The accident occurred on January 10, 1992; the toys floated for years.
Sequence	The particular order in which a series of events or processes occurs.	What order?	Containers lost, container broke open, toys released, toys floated, toys made landfall.

Part 2: Scientific Models and Data Analysis (8 terms)

Term	Definition	Think:	Example
Model	A representation of an object, process, or system that scientists use to understand, explain, or predict how something works.	How is it being represented ?	Computer-based global climate models represent future Earth conditions.
Pattern	A recognizable trend, repetition, or regular way that something happens or changes.	What trend do I notice?	Greenhouse gas concentration in Scenario A increases over time from 2000-2100.
Relationship	The way two or more variables are connected or how they change in relation to one another.	How are they connected?	As greenhouse gas concentration increases, global surface temperature generally increases.
Projected	Predicted or estimated to happen in the future based on available information or data.	What is expected to happen in the future?	Future temperature values on the graph are projected, not observed.
Distribution	The way values or information are spread out across an area, range, or set of categories.	How are the values spread out?	The four scenarios' 2100 values are spread across a wide range.
Anomaly	A difference or change from an expected or historical average value.	How different is it from the expected or average value?	An anomaly of 0 degrees C means no change from the historical average; a positive anomaly means warmer than average.
Claim	A statement or conclusion that something is true and can be supported with evidence.	What am I saying is true?	"Earth's global surface temperature increased from 1920 to 2020" is a claim.
Evidence	Information, observations, or data used to support a claim or explanation.	How do I know?	The specific temperature anomaly values from the graph (about -0.2 degrees C in 1920 to about 1.0 degrees C in 2020) are the evidence.

Part 3: Features, Systems, and Engineering (12 terms)

Term	Definition	Think:	Example
Feature	A distinctive or noticeable characteristic, part, or aspect of something.	What characteristic or part do I notice?	Solution C uses earthen barriers to dam the lava.
System	A group of interacting or connected parts that work together as a whole.	What parts are working together?	Lava, land, people, buildings, equipment, and proposed solutions all affect one another.
Component	One individual part of a larger system.	What is one part of the whole?	Lava flow, pumps, community, and barriers are all components of the lava-flow management system.
Solution	A method or action designed to solve a problem or address a challenge.	How could the problem be addressed?	Lava diversion, seawater cooling, lava flow barriers, and explosions are all proposed solutions.
Criteria	The requirements or desired characteristics that a successful solution should meet.	What does the solution need to do?	A successful lava solution must protect homes, reduce lava damage, and be reliable during an eruption.
Constraint	A limitation or restriction that must be considered when designing or choosing a solution.	What limits my options?	Seawater cooling is constrained by the need for a constant water supply and energy for pumps.
Trade-off	A compromise in which gaining a benefit in one area may result in a disadvantage or cost in another.	What do I gain, and what do I give up?	Seawater cooling helps cool and harden lava, but it requires a constant water supply and costly energy.
Mitigation	An action taken to reduce the severity, impact, or harmful effects of a problem.	How can the impact be reduced?	Redirecting a lava flow doesn't stop the eruption, but it reduces the damage it causes.
Simulation	An imitation or model of a real-world situation or process used to investigate what may happen under different conditions.	What happens if...?	The EN-ROADS simulator lets users change variables and observe possible effects on climate.
Input	Information, conditions, matter, or energy that enters a system or is put into a model.	What goes in?	Rate of tree planting, deforestation, greenhouse gas emissions, and population growth are inputs to the simulation.
Output	The information, result, matter, or energy that comes out of a system or model.	What comes out?	The projected temperature increase (for example, +2.5 degrees C by 2100) is the output/result.
Feedback	A process in which a change in one part of a system causes effects that can influence other parts of the system, including the original change.	How can one change lead to more change?	Increased human activity leads to more greenhouse gases, which leads to rising temperatures, which leads to even more emissions.